Where the BFKL terms come from, and how the mid-rapidity result compares with the forward one (arXiv:1411.2869)

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1 Summary

Notation.

- $\mathcal{J} \equiv$ JIMWLK $\otimes$ LO: your App. F, the bracket of Sec. 3 of `Evolution_terms_corrected.pdf`.
- $\mathcal{B} \equiv$ BFKL $\otimes$ LO: your App. E, the bracket of its Sec. 4.
- $L_T \equiv \log \frac{k^+}{\Lambda}$ (target side), $L_P \equiv \log \frac{\vee}{k^+}$ (projectile side), $L_T + L_P = \log \frac{\vee}{\Lambda}$.

1. Your input terms ($\propto \log \frac{\vee}{\Lambda}$) give JIMWLK only.

- All terms of your `Evolution_terms.tex`, Secs. 1 and 2 (with the `4rho.tex` lists) add up exactly to $\log \frac{\vee}{\Lambda}\, \mathcal{J}$. This was checked for a general adjoint U (`Evolution_terms_check.pdf`).
- They cannot contain $\mathcal{B}$:
  - in every input term (474 in Secs. 1, 2 and 30 in the `4rho.tex` lists) the measured gluon is emitted by a valence charge, with a kernel $\mathcal{K}(\text{charge} - w) \cdot \mathcal{K}(\text{charge}' - w')$;
  - in 23 of the 35 terms of $\mathcal{B}$ it is emitted by the soft gluon at z, e.g. $\mathcal{K}(w - z) \cdot \mathcal{K}(w' - z)$.

  No sum of the input terms produces those 23.

2. The BFKL terms are in the $\log \frac{\vee}{k^+}$ terms you put among the finite ones.

- Every row is a p^+ integral over $(\Lambda, \vee)$. Its energy denominator splits $\frac{1}{p^+}$ into a part that lives at $p^+ \ll k^+$ and a part that lives at $p^+ \gg k^+$ (Sec. 2, Fig. 1A).
- The first gives L_T, the second L_P. Your rule $\log \frac{k^+}{\Lambda} = \log \frac{\vee}{\Lambda} - \log \frac{\vee}{k^+}$ moves the first into $\log \frac{\vee}{\Lambda}$.

- The terms in which the soft gluon emits the measured gluon exist only for $p^+ > k^+$. So they can only appear with $\log \frac{\vee}{k^+}$.
- All rows together (checked row by row in `LogVeeKplus_BFKL.pdf`):

$$\left.\frac{d^3N}{d^3k}\right|_{\rm LL} = \log\frac{\vee}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J}) = L_T\,\mathcal{J} + L_P\,\mathcal{B}. \tag{1.1}$$

3. No interpretation is needed. The two equations follow from the calculation, by requiring that the cross section does not depend on the two cutoffs Λ and $\vee$ (Sec. 3). This is the method AABKL use to derive BK, their Eqs. (5.32), (5.34).

- Λ **(target side).** Changing Λ must be compensated by changing the target. The response is the coefficient of L_T, i.e. $\mathcal{J}$. It acts only on the Wilson lines: this is JIMWLK.
- $\vee$ **(projectile side), at fixed** k^+**.** Changing $\vee$ must be compensated by changing the charge distribution of the projectile. The response is the coefficient of L_P, i.e. $\mathcal{B}$. It acts only on $\langle\rho\rho\rangle$: this is BFKL.

4. The coefficient of $\log\frac{\vee}{\Lambda}$ **alone has no meaning.** Only the coefficients of L_T and L_P are unique.

- With your rule (eliminate L_T) the coefficient of $\log\frac{\vee}{\Lambda}$ is $\mathcal{J}$.
- Eliminating L_P instead makes it $\mathcal{B}$: $L_T\mathcal{J} + L_P\mathcal{B} = \log\frac{\vee}{\Lambda}\,\mathcal{B} + L_T(\mathcal{J}-\mathcal{B})$.
- So "JIMWLK has $\log\frac{\vee}{\Lambda}$" holds only in your bookkeeping. The statement that holds in any bookkeeping is "JIMWLK with $\log\frac{k^+}{\Lambda}$, BFKL with $\log\frac{\vee}{k^+}$". At mid rapidity both are $\frac{1}{2}\log\frac{\vee}{\Lambda}$.

5. Forward (AABKL) vs mid rapidity (Sec. 4). Your collaborator is right for the target side and the final state, and not for the projectile side.

- **The rapidity split is the same.**
  - AABKL evolve the projectile only over $Y_P = \ln\frac{1}{x_p} + O(1)$ and the target over Y_T (their Eq. (2.11) and Fig. 1). Their $x_p = p^+/P^+$ is our $k^+/\vee$, so $\ln\frac{1}{x_p} = L_P$.
  - At forward rapidity $x_p \sim 1$: $L_P = O(1)$, so there is no projectile evolution, and they derive BK for the target at fixed x_p.
  - Eq. (1.1) at fixed $k^+/\vee$ gives exactly this.
  - They also note that for small x_p the projectile distribution depends on P^+ (their Sec. III.A, item 3). That is the mid-rapidity situation, where $L_P = \frac{1}{2}\log\frac{\vee}{\Lambda}$ must be resummed: BFKL.
- **Same:**
  - the target evolution (BK/JIMWLK, same kernel and colour structures);
  - the final-state side: the fragmentation DGLAP term with the same P_{gg}, the same plus prescription, the same real-emission limit and the same scheme constant ($c_0^2 = 1.26$);
  - the hadron convolution with lower limit $x_F \leftrightarrow \frac{k^+}{\vee}$.
- **Different, and not a small change:**
  - LO itself: a collinear parton times a dipole, $O(\alpha_s^0)$, vs. a gluon radiated by the charges, $O(\alpha_s)$;
  - hence the projectile side: BFKL over L_P (absent forward), no PDF DGLAP, running coupling of the LO coupling, linear polarization, emission after the target by the rotated charge, three- and four-ρ terms;

– the regularization of the soft gluon: AABKL's Ioffe-time cutoff vs. our sharp p^+ cutoffs. This moves $O(\alpha_s)$ terms between "evolution" and "finite" (their Eqs. (3.11)–(3.13)).

- **One choice to make:** AABKL's virtual term has no x_p limit ($\int_0^1 d\xi$). With that choice the $\log^2(1-\frac{k^+}{\vee})$ term and the Q1 pole of the previous note both disappear (option (ii) of `Finite_terms_singlet.pdf`, Sec. 2).

2 How a p^+ integral splits

General statement. Each row is, before any rewriting, an integral over the soft-gluon momentum,

$$\int_\Lambda^\vee \frac{dp^+}{p^+}\, G\Big(\frac{p^+}{k^+}\Big), \qquad G_T \equiv \lim_{p^+\ll k^+} G, \qquad G_P \equiv \lim_{p^+\gg k^+} G. \tag{2.1}$$

- G depends on p^+ only through p^+/k^+ (boost invariance of the energy denominators). So the integral has two logarithms, one from each end:

$$\int_\Lambda^\vee \frac{dp^+}{p^+}\, G = G_T\, \log\frac{k^+}{\Lambda} + G_P\, \log\frac{\vee}{k^+} + \text{finite}. \tag{2.2}$$

- With your rule $\log\frac{k^+}{\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}$:

$$= G_T\, \log\frac{\vee}{\Lambda} + \big(G_P - G_T\big) \log\frac{\vee}{k^+} + \text{finite}. \tag{2.3}$$

- So your $\log\frac{\vee}{\Lambda}$ lists contain only G_T: the soft gluon slower than the measured one. Everything that happens only for $p^+ > k^+$ is in G_P, and therefore only in the $\log\frac{\vee}{k^+}$ terms.

Explicit example: the energy denominator of your rows. In Groups II and III the denominator is $D = p^+A + k^+B$, with $A = (x-z)^2$ and $B = (x-w)^2$. Exactly,

$$\frac{1}{p^+} = \frac{k^+B}{p^+D} + \frac{A}{D}, \tag{2.4}$$

$$\int_\Lambda^\vee dp^+\, \frac{k^+B}{p^+(p^+A + k^+B)} = \log\frac{k^+}{\Lambda} + \log\frac{B}{A} + O\Big(\frac{\Lambda}{k^+}, \frac{k^+}{\vee}\Big) \qquad (G_T = 1,\ G_P = 0), \tag{2.5}$$

$$\int_\Lambda^\vee dp^+\, \frac{A}{p^+A + k^+B} = \log\frac{\vee}{k^+} + \log\frac{A}{B} + O\Big(\frac{\Lambda}{k^+}, \frac{k^+}{\vee}\Big) \qquad (G_T = 0,\ G_P = 1). \tag{2.6}$$

- The two pieces add up to $\log\frac{\vee}{\Lambda}$. The switch happens at $p^+ \approx k^+B/A$, i.e. at $p^+ \sim k^+$; the transverse ratio gives only the finite $\log\frac{A}{B}$ (Fig. 1A).

- The second piece is where the soft gluon is faster than the measured one. The measured gluon can then be emitted by it: these are the BFKL terms with $\mathcal{K}(w-z)$, $\mathcal{K}(w'-z)$. They have $G_T = 0$, so they never appear with $\log\frac{k^+}{\Lambda}$, and with your rule never with $\log\frac{\vee}{\Lambda}$.

- In some rows the same split happens through a phase instead: Group III Rows III, V, VI give $E_1(i\kappa) - E_1(i\kappa/a) \to \log\frac{\vee}{k^+}$ only when the measured gluon sits at the soft gluon ($y' \to z$).

Sum over all rows. $\sum_{\text{rows}} G_T = \mathcal{J}$ and $\sum_{\text{rows}} G_P = \mathcal{B}$, which gives Eq. (1.1). Both sums were checked earlier:

- the $\log\frac{\vee}{\Lambda}$ terms: `Evolution_terms_check.pdf` (exact, general U);

- the $\log\frac{\vee}{k^+}$ terms: `LogVeeKplus_BFKL.pdf`, equal to $\mathcal{B} - \mathcal{J}$ in all 18 kernel structures, pointwise in z. In the singlet projection $\mathcal{B}$ is your App. E.

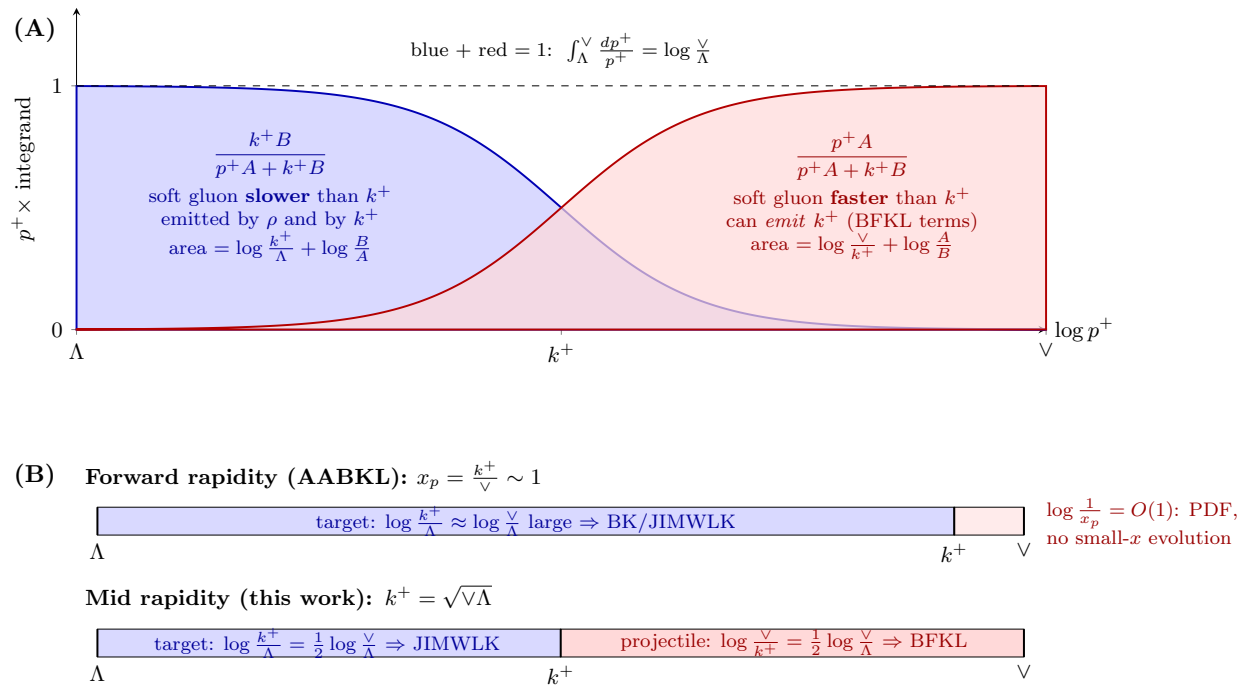


Figure 1: (A) The p^+ integrand of a row, times p^+, against $\log p^+$. The energy denominator splits it into a part at $p^+ < k^+$ (blue: soft gluon slower, emitted by ρ and by the measured gluon, target side) and a part at $p^+ > k^+$ (red: soft gluon faster, can emit the measured gluon, projectile side). The areas are $\log\frac{k^+}{\Lambda}$ and $\log\frac{\vee}{k^+}$ (plus $\pm\log\frac{A}{B}$). (B) Forward rapidity: the projectile interval $\log\frac{1}{x_p}$ is $O(1)$, only the target evolves (AABKL). Mid rapidity: both intervals are $\frac{1}{2}\log\frac{\vee}{\Lambda}$, both evolve.

3 Why this is not an interpretation: independence of the cutoffs

Λ and $\vee$ are not physical. Λ separates the soft gluons from the target fields; $\vee$ separates them from the valence charges ρ. The physical cross section cannot depend on where the two boundaries are put. At this order the dependence enters only through L_T and L_P in Eq. (1.1):

$$-\Lambda\frac{\partial}{\partial\Lambda}\frac{d^3N}{d^3k}\bigg|_{\vee,k^+} = \mathcal{J} = -H_{\rm JIMWLK}\,\frac{d^3N_{\rm LO}}{d^3k}, \tag{3.1}$$

$$\vee\frac{\partial}{\partial\vee}\frac{d^3N}{d^3k}\bigg|_{\Lambda,k^+} = \mathcal{J} + (\mathcal{B}-\mathcal{J}) = \mathcal{B} = {\rm LO}\Big[\langle\rho\rho\rangle \to K_{\rm BFKL}\otimes\langle\rho\rho\rangle\Big]. \tag{3.2}$$

- **Eq.** (3.1)**.** Lowering Λ moves gluons from the target into the soft window. To compensate, the target's Wilson-line distribution must change by $\mathcal{J}$. $\mathcal{J}$ is $-H_{\rm JIMWLK}$ acting on the Wilson lines of LO; the charges are untouched. This is the JIMWLK equation, here derived, not assumed.

- **Eq.** (3.2)**.** Raising $\vee$ moves charges from ρ into the soft window. To compensate, the projectile's charge correlator must change by $\mathcal{B}$. $\mathcal{B}$ is LO with $\langle\rho^f(x)\rho^f(y)\rangle$ replaced by its coordinate-space BFKL evolution, and the Wilson lines are untouched. This is the BFKL equation.

- **If the** $\log\frac{\vee}{\Lambda}$ **terms contained** $\mathcal{B}$, Eq. (3.1) would require a change of the target by $\mathcal{J}+\mathcal{B}$. No change of the target Wilson lines can produce the 23 terms of $\mathcal{B}$ in which the measured gluon is emitted by the soft gluon. So the structure of Eq. (1.1) is forced.

- **The same logic in AABKL.** They write: "we do not have to specify what is exactly the evolution equation ... This equation is self-consistently determined from the calculation itself" (their Sec. II.A). They then derive BK from $s_0\frac{d}{ds_0}\,d\sigma = 0$ (their Eq. (5.34)). Their s_0 is their target-side cutoff (Ioffe time), the analogue of our Λ, so this is Eq. (3.1).

- **Wave-function picture** (`NLO_gluon_midrapidity.pdf`, Sec. 6). At leading log, the emissions are ordered in p^+, the harder first:
  - for $p^+ < k^+$, $\Omega = \Omega_p[\rho + j_k]\,\Omega_k[\rho]$: the soft gluon is emitted by ρ and by the measured gluon. This gives JIMWLK;
  - for $p^+ > k^+$, $\Omega = \Omega_k[\rho + j_p]\,\Omega_p[\rho]$: the measured gluon is emitted by ρ and by the soft gluon. This gives BFKL.

At mid rapidity, $L_T = L_P = \frac{1}{2}\log\frac{\vee}{\Lambda}$, so Eq. (1.1) is $\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B})$, the form of your corrected evolution file. Away from mid rapidity it is $\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B}) + \log\frac{k^+}{\sqrt{\vee\Lambda}}(\mathcal{J}-\mathcal{B})$.

4 Forward rapidity (AABKL) vs mid rapidity (this calculation)

4.1 Setups

- **AABKL** (T. Altinoluk, N. Armesto, G. Beuf, A. Kovner, M. Lublinsky, arXiv:1411.2869): hybrid formalism.
  - The projectile is a collinear proton (PDFs); the produced parton has large $x_p = p^+/P^+$.
  - LO is $\frac{1}{(2\pi)^2} x_p f(x_p) \int e^{ip(y-\bar{y})} s_{Y_T}[y,\bar{y}]$, of order α_s^0.
  - NLO is one extra gluon in the projectile parton's wave function.
  - The soft-gluon phase space is cut by an Ioffe-time condition, $\xi(1-\xi)x_p P^+/k_\perp^2 > \tau$.
- **This calculation:** the projectile is dilute valence charges ρ with momentum $\vee$.
  - LO is the gluon radiated by ρ (Kovchegov–Tuchin type), of order α_s, with $\langle\rho\rho\rangle$ and the trace $\mathrm{Tr}\{(U^\dagger(x) - U^\dagger(y))(U(x') - U(y'))\}$.
  - NLO is one extra gluon in the window $\Lambda < p^+ < \vee$, with sharp cutoffs.

4.2 Rapidity intervals and evolution

- AABKL, Sec. II.A: "the projectile wave function at any energy is evolved only to rapidity $Y_p = \ln\frac{1}{x_p} + Y_0$"; the target is evolved by Y_T (their Eq. (2.11), Fig. 1).
  - With $x_p \leftrightarrow \frac{k^+}{\vee}$: $\ln\frac{1}{x_p} = L_P$, and Y_T plays the role of L_T.
  - This is the split of Eq. (1.1).
- They derive the energy dependence at *fixed* x_p (their Eq. (5.32): "production at fixed x_p but at higher energy"). At fixed $a = \frac{k^+}{\vee}$, $L_P = \log\frac{1}{a}$ is constant and Eq. (1.1) gives $\frac{d}{d\log\vee}\frac{d^3N}{d^3k} = \mathcal{J}$: only the target evolves. This is their BK result. Their Eq. (5.39) is the same evolution acting on a fundamental dipole, i.e. the quark channel.
- They neglect the P^+ dependence of the projectile distribution because x_p is not small. In their words (Sec. III.A, item 3): "For small values of x one indeed has to take into account the P^+ dependence".
  - At mid rapidity $x_p = \frac{k^+}{\vee} = \sqrt{\Lambda/\vee}$ is small and $L_P = \frac{1}{2}\log\frac{\vee}{\Lambda}$ is as large as the target interval.
  - That P^+ (i.e. $\vee$) dependence is our Eq. (3.2): the BFKL evolution of the projectile.
- **So at leading log, the forward limit of our result is AABKL's structure.** The one difference at mid rapidity is that the projectile interval is large and must be resummed.

4.3 Term by term

AABKL (forward), Eqs. (4.20)–(4.22)	this work (mid rapidity), `Finite_terms_singlet.pdf` Eq. (7.1)		comment
target evolution: BK for s_{Y_T}, from s_0-independence	$L_T\,\mathcal{J}$ (JIMWLK), from Λ-independence	same	same kernel; BK is the large-N_c dipole form
projectile: PDF $f(x_p)$, no small-x evolution	$L_P\,\mathcal{B}$ (BFKL of $\langle\rho\rho\rangle$)	different	$L_P = \ln\frac{1}{x_p}$ is $O(1)$ forward, $\frac{1}{2}\log\frac{\vee}{\Lambda}$ at mid rapidity
LO: $x_p f(x_p)\, s_{Y_T}[y,\bar{y}]$, $O(\alpha_s^0)$	LO: WW gluon of ρ, $O(\alpha_s)$, $\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(y))(U(x')-U(y'))\}$	different	collinear parton vs. $k_\perp$-dependent emission by colour charges
hadron convolution $\int_{x_F}^1 \frac{d\zeta}{\zeta^2} D(\zeta)\ldots$	$\int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^2} D_{h/g}(\zeta)\ldots$	same	lower limit from the largest projectile momentum
fragmentation (after-target) term, P_{gg} with $[\cdot]_+$, $\int_0^{1-x_p} d\xi$	DGLAP term, $-\frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}\mathrm{LO}_\xi[\ell_\Delta]$	same	same splitting function, plus prescription and limit; $f_{\mu^2} = f^{\overline{MS}}_{1.26\mu^2}$ is our $c_0^2 = 1.26$
PDF (before-target) collinear term	G2 Row I part 2: finite	different	our parent is emitted at distance $\lvert x-w\rvert$ from a charge; no collinear pole, no PDF
interference and virtual colour structures (sss, $\overline{X}/N_c^2$)	$\mathrm{Tr}\{T^e A T^e B\}$ traces of Secs. 5, 6	same colour	kinematics differ (emission by charges, energy denominators $p^+A + k^+B$)
virtual term, $\int_0^1 d\xi$ (no x_p limit)	G1 Row II: limit $\xi \geq \frac{k^+}{\vee} \Rightarrow V_\vee$ and the Q1 pole	different	with AABKL's choice both disappear
soft gluon cut by the Ioffe time; "evolution interval" depends on $l_\perp$ (their Eq. (3.13))	sharp cutoffs $\Lambda < p^+ < \vee$	scheme	changes $O(\alpha_s)$ finite terms, not the logarithms
—	running coupling of the LO g^2; CMW constant $\frac{67}{9} - \frac{\pi^2}{3}$	only here	AABKL's LO has no coupling to renormalize
—	linear polarization term	only here	our incoming gluon is the polarized WW field
—	the LO (parent) gluon emitted after the target by the rotated charge	only here	charges, not a parton, radiate; AABKL's parent parton exists before the target
—	three- and four-ρ terms	only here	several charges; zero or Gaussian-reducible for a Gaussian projectile
quark channels (C_F terms, Eq. (4.21))	—	only there	$n_f = 0$, no quarks in our projectile

4.4 Verdict

- **"Mostly the same" holds for:** the target evolution, the final-state (fragmentation) side, the hadron convolution, and the colour structures of the one-gluon real and virtual terms. These are the parts that do not depend on what the projectile is.
- **It does not hold for the projectile side**, because the two LOs are different objects: a collinear parton with a PDF vs. a gluon radiated by colour charges. This difference is not a small correction:
  - at mid rapidity there is a second large logarithm, $\log\frac{\vee}{k^+}$, with BFKL;

  - there are the running coupling of LO, the linear polarization, emission after the target, and the multi-charge terms;
  - there is no PDF DGLAP.

- **The finite constants are scheme dependent** (Ioffe time vs. sharp cutoffs). A term-by-term numerical agreement of the finite parts is not expected even where the structures match.

- **Consistency check that does hold:** at leading log, our Eq. (1.1) at fixed $k^+/\vee$ reproduces AABKL's evolution structure: target only, interval set by the produced particle, nothing for the projectile.